



## PROJECT ON:-

### Mini Banking Transaction Logger

|                     |                         |
|---------------------|-------------------------|
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| Slot                | B21+B22+B23+E14+E21+E22 |

# Introduction:

This project is a simple console-based banking transaction logger. It allows a user to record deposits and withdrawals, check the current balance, view transaction history and delete any wrong entry. The main aim of making this project is to practice the basic Python concepts taught in the course such as lists, dictionaries, functions, loops and conditionals. The program is completely menu driven and easy to use.

## Problem Statement:

People often need to keep a record of their daily money transactions. Writing everything on paper is not convenient and calculating the balance manually can lead to mistakes. There is a need for a simple program that can store deposits and withdrawals, show the current balance and give a clear summary of all transactions. This project tries to solve that problem using only the basic programming concepts learned in class.

## Objectives:

- To create a simple program that can add deposit and withdrawal entries
- To calculate and display the current balance correctly
- To show all transactions and also filter them by type
- To allow the user to delete any transaction if needed
- To practice using lists, dictionaries, functions and loops

# Functional Requirements:

The project has three main modules:

## 1. Transaction Management Module

- Add a new deposit
- Add a new withdrawal
- Delete a transaction using its ID

## 2. Calculation Module

- Calculate current balance
- Calculate total deposits
- Calculate total withdrawals

## 3. Reporting Module

- Display all transactions
- Show only deposits
- Show only withdrawals
- Show account summary

# Non-Functional Requirements:

1. **Usability** – The menu is simple and easy to understand.
2. **Reliability** – The program checks for invalid amounts and shows proper messages.
3. **Maintainability** – The code is divided into different functions so it is easy to understand and change.
4. **Performance** – The program works quickly even with many transactions because it uses simple list operations.
5. **Error Handling** – Wrong inputs like negative amounts or invalid IDs are handled properly.

# System Architecture:

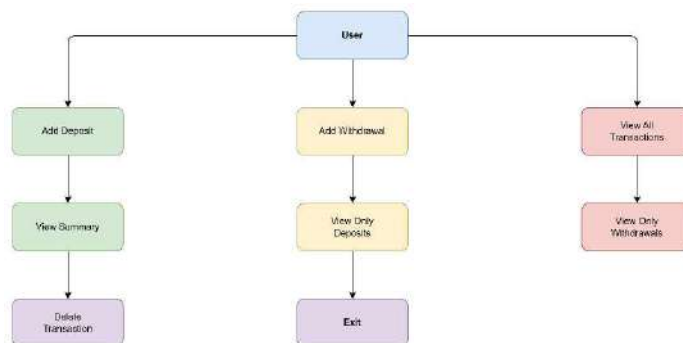
The program is divided into different parts:

- `main.py` controls the menu and overall flow
- `transactionmanager.py` handles adding and deleting transactions
- `calculator.py` does all the balance and total calculations
- `reporter.py` shows the transactions and summary
- `datahandler.py` helps in finding a transaction and creating new IDs.

## Design Diagrams:

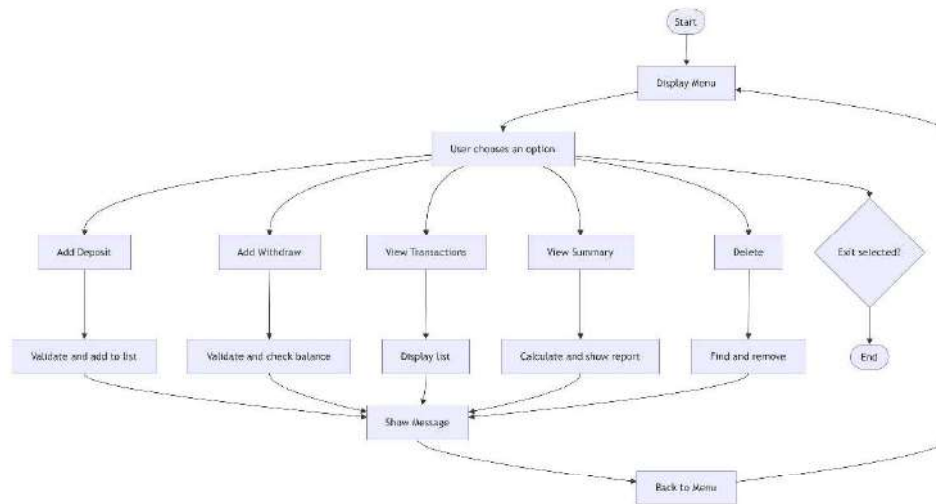
### Use Case Diagram

The main actor is the User. The user can add deposit, add withdrawal, view transactions, view summary, view only deposits, view only withdrawals and delete a transaction.



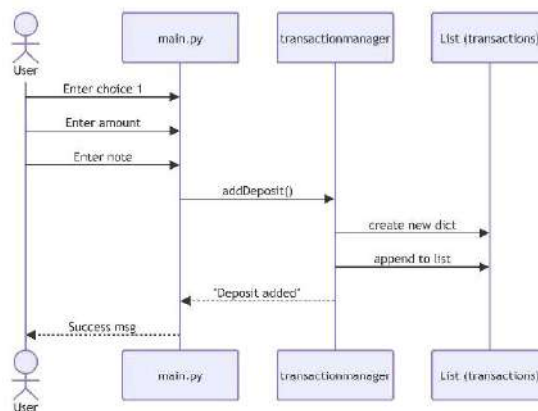
### Workflow Diagram

The program starts and shows the menu. The user selects an option. Based on the choice the related function is called. After the operation is completed the menu is shown again until the user chooses to exit.



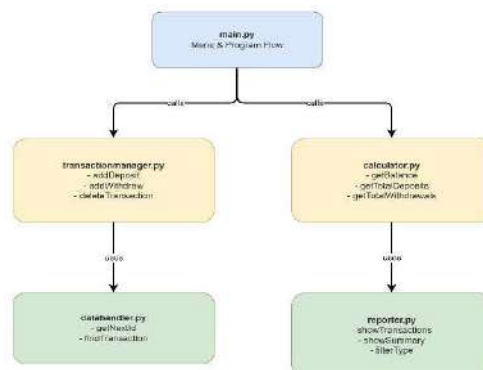
## Sequence Diagram (for Add Deposit)

User enters the amount and note → main function calls addDeposit → new transaction is created and added to the list → success message is shown to the user.



## Component Diagram

The main components are Main Controller, Transaction Manager, Calculator and Reporter. They work together to complete the tasks.



# Design Decisions and Rationale:

- Console based program was chosen because it matches the level of the course and does not need any extra libraries.
- List of dictionaries is used to store transactions because it is easy to understand and taught in the syllabus.
- Separate functions are used for different tasks so that the code remains clean and modular.
- No file storage is used to keep the project simple and within the basic concepts.

## Implementation Details:

The main data structure is a list. Each transaction is stored as a dictionary with keys like id, type, amount and note.

Functions are created for:

- Adding deposit and withdrawal
- Calculating balance and totals
- Displaying transactions
- Filtering by type
- Deleting a transaction

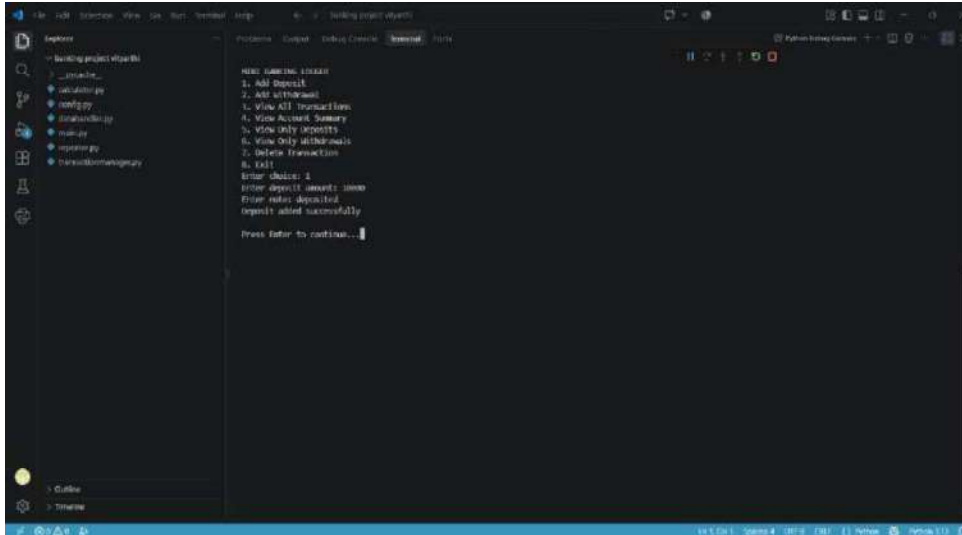
## Testing Approach:

Testing was done manually by performing the following steps:

- Added some deposit and withdrawal entries
- Checked if the balance is calculated correctly
- Viewed only deposits and only withdrawals
- Deleted a transaction and confirmed it was removed
- Entered wrong values like negative amount and invalid ID to check error messages
- Checked the summary to confirm totals are correct

# Screenshots / Results:

## 1.Add Deposit



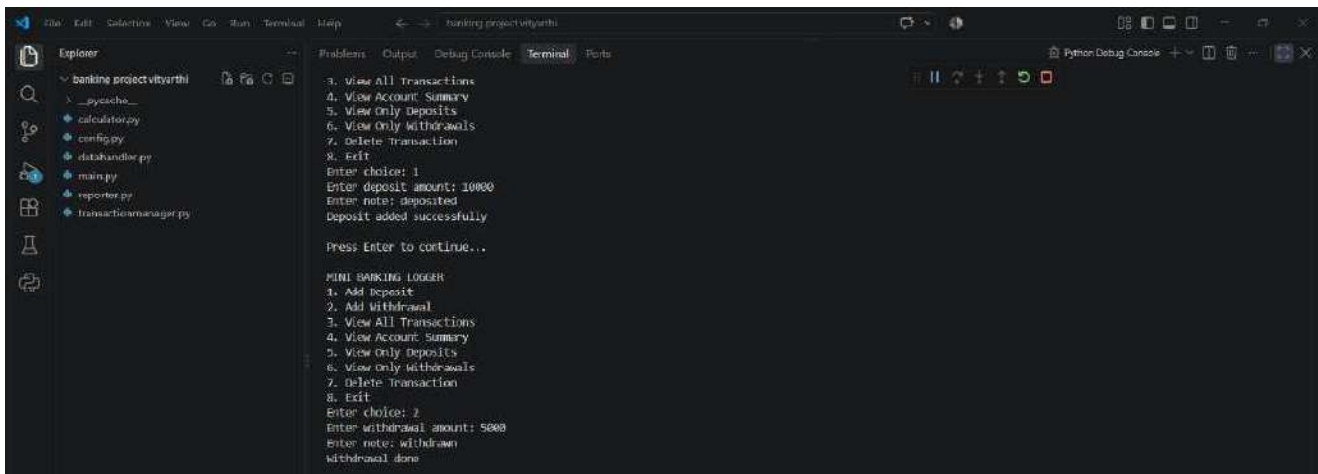
```
banking project vjyarthi
> __main__
> calculator.py
> config.py
> datahandler.py
> main.py
> reporter.py
> transactionmanager.py

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 1
Enter deposit amount: 10000
Enter note: deposited
Deposit added successfully

Press Enter to continue...
```

## 2.Add Withdrawal



```
banking project vjyarthi
> __pycache__
> calculator.py
> config.py
> datahandler.py
> main.py
> reporter.py
> transactionmanager.py

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Python Debug Console

1. View All Transactions
2. Add Withdrawal
3. View Account Summary
4. View Only Deposits
5. View Only Withdrawals
6. Delete Transaction
7. Exit
Enter choice: 1
Enter deposit amount: 10000
Enter note: deposited
Deposit added successfully

Press Enter to continue...

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 2
Enter withdrawal amount: 5000
Enter note: withdrawn
Withdrawal done
```

## 3.View All Transactions

```
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 2
Enter withdrawal amount: 5000
Enter note: withdraw
Withdrawal done

Press enter to continue...

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 3

ID      Type      Amount      Note
-----
1  Deposit  10000.0    deposited
2  Withdraw  5000.0     withdraw

Press enter to continue...

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 1
```

| ID | Type     | Amount  | Note      |
|----|----------|---------|-----------|
| 1  | Deposit  | 10000.0 | deposited |
| 2  | Withdraw | 5000.0  | withdraw  |

4.View Account Summary

```
Enter choice: 3

ID      Type      Amount      Note
-----
1  Deposit  10000.0    deposited
2  Withdraw  5000.0     withdraw

Press enter to continue...

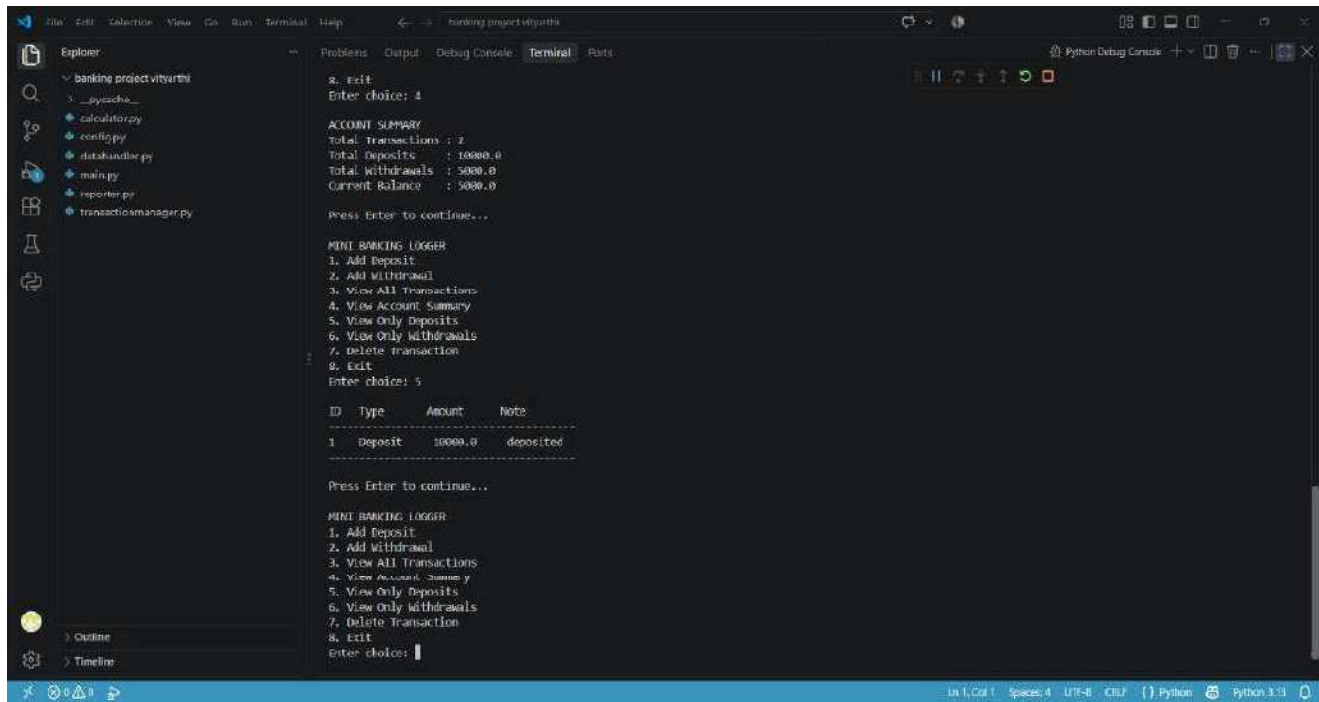
MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 4

ACCOUNT SUMMARY
Total Transactions : 2
Total Deposits    : 10000.0
Total Withdrawals : 5000.0
Current Balance   : 5000.0

Press enter to continue...

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 1
```

5.View Only Deposits



```
8. Exit
Enter choice: 4

ACCOUNT SUMMARY
Total Transactions : 2
Total Deposits : 10000.0
Total Withdrawals : 5000.0
Current Balance : 5000.0

Press Enter to continue...

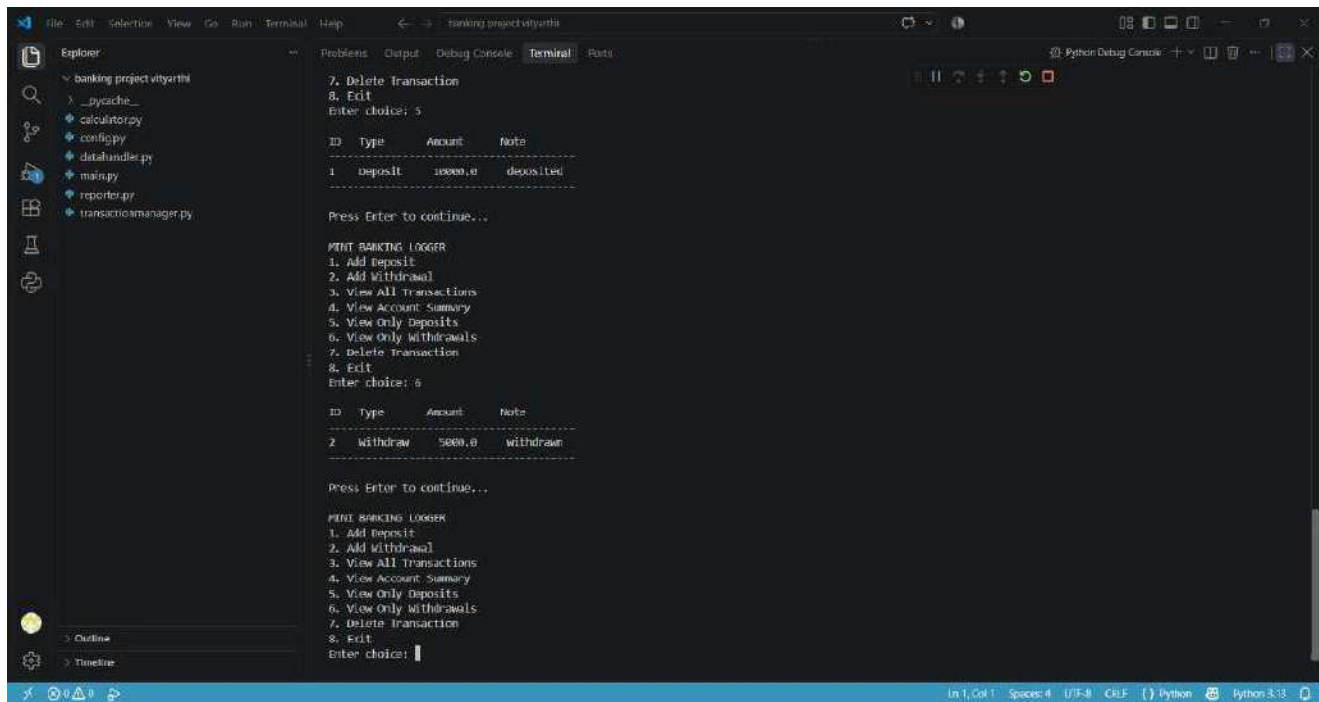
MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete transaction
8. Exit
Enter choice: 5

ID Type Amount Note
-----
1 Deposit 10000.0 deposited

Press Enter to continue...

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete transaction
8. Exit
Enter choice: |
```

## 6.View Only Withdrawals



```
7. Delete transaction
8. Exit
Enter choice: 5

ID Type Amount Note
-----
1 Deposit 10000.0 deposited

Press Enter to continue...

MINI BANKING LOGGER
1. Add deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete transaction
8. Exit
Enter choice: 6

ID Type Amount Note
-----
2 Withdraw 5000.0 withdrawn

Press Enter to continue...

MINI BANKING LOGGER
1. Add deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete transaction
8. Exit
Enter choice: |
```

## 7.Delete Transaction

```
File Edit Selection View Go Run Terminal Help
banking project vityarthis

Explorer
banking project vityarthis
  __pycache__
  calculator.py
  config.py
  datahandler.py
  main.py
  reporter.py
  transactionmanager.py

Problems Output Debug Console Terminal Ports
Python Debug Console

ID Type Amount Note
-----
2 Withdraw 5000.0 withdrawn

Press Enter to continue...

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 7

ID Type Amount Note
-----
1 Deposit 10000.0 deposited
2 Withdraw 5000.0 withdrawn

Enter ID to delete: 2
Transaction deleted

Press Enter to continue...

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 
```

## 8.Exit

```
File Edit Selection View Go Run Terminal Help
banking project vityarthis

Explorer
banking project vityarthis
  __pycache__
  calculator.py
  config.py
  datahandler.py
  main.py
  reporter.py
  transactionmanager.py

Problems Output Debug Console Terminal Ports
Python Debug Console

2 Withdraw 5000.0 withdrawn

Press Enter to continue...

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 7

ID Type Amount Note
-----
1 Deposit 10000.0 deposited
2 Withdraw 5000.0 withdrawn

Enter ID to delete: 2
Transaction deleted

Press Enter to continue...

MINI BANKING LOGGER
1. Add Deposit
2. Add Withdrawal
3. View All Transactions
4. View Account Summary
5. View Only Deposits
6. View Only Withdrawals
7. Delete Transaction
8. Exit
Enter choice: 8
Thank you, Bye!
PS C:\banking project vityarthis>
```

## Challenges Faced:

One small challenge was handling the case when the user tries to withdraw more money than the available balance. This was solved by checking the current balance before allowing the withdrawal. Another small issue was making sure the transaction ID is unique. This was solved by finding the highest existing ID and adding one to it.

## Learnings and Key Takeaways:

Through this project I learned how to break a problem into smaller parts and write separate functions for each part. I also got better practice in using lists and dictionaries. Working with a menu-driven program helped me understand how real programs take input from the user and respond accordingly. Overall it improved my confidence in writing modular Python code.

## Future Enhancements:

- Add a feature to search transaction by note
- Allow editing of an existing transaction
- Show transactions in sorted order by amount
- Add a simple date for each transaction

# References:

1. Course materials of CSE1021 – Introduction to Problem Solving and Programming
2. Python official documentation (basic data structures)
3. Class notes on lists, dictionaries, functions and control flow
4. VITyarthi course (Python Essentials by Dr. Laxmi D.)
5. Python Programming using problem solving approach (By Reema Thareja)